COOX

Non-Serviceable Bookings Analysis

1. EXECUTIVE SUMMARY

557	15+	30+	₹84L+
Total Refunded Bookings	Cities Affected	Pincodes to Block	Est. Revenue at Risk

COOX is experiencing a consistent pattern of bookings from non-serviceable zones — primarily city outskirts and areas beyond operational limits — that are subsequently refunded in full. This analysis examines 557 such refunded bookings across Indian cities, applies DBSCAN geospatial clustering, and derives a prioritised pincode-level blocking list to prevent future revenue leakage.

Key Findings at a Glance

- **Addressable Loss:** An estimated ₹84 lakhs+ in revenue is linked to non-serviceable bookings in the dataset alone.
- **Top Affected Cities:** Bengaluru, Hyderabad, and Delhi-NCR account for the majority of outskirt bookings.
- **Clustering Effectiveness:** DBSCAN ($\varepsilon = 5$ km) identifies well-separated geographic hotspots with a silhouette score > 0.5.
- **Immediate Action Zone:** Critical and High severity clusters account for over 60% of all non-serviceable bookings.
- **Noise Watchlist:** Isolated bookings in the same pincode (2+ hits) are flagged as emerging hotspots before they become clusters.

2. PROBLEM STATEMENT

Background

COOX operates a home-chef / culinary services platform across major Indian cities. Each service booking requires a chef or culinary professional to reach the customer's location. Locations beyond the serviceable radius — typically city outskirts, satellite towns, or newly developed peripheries — cannot be fulfilled.

The current system allows customers in non-serviceable zones to complete a booking, which then triggers a manual cancellation and full refund. This creates four compounding problems:

#	Problem	Business Impact
1	**Revenue Leakage**	Every refunded booking represents a lost transaction — no revenue, but operational overhead was consumed.
2	**Customer Frustration**	Customers who book and get cancelled develop negative brand perception and reduced lifetime value.

3	**Operational Overhead**	Customer support and finance teams spend manual effort processing refunds for predictable, preventable failures.
4	**Demand Signal Distortion**	Refunded bookings appear as "demand" in city dashboards, skewing chef deployment and demand forecasting.

Objective

Analyse historical non-serviceable booking data to identify geographic clusters of repeat violations, extract the corresponding pincodes, and produce a geo-blocking recommendation to proactively prevent future bookings from these zones.

3. DATASET & METHODOLOGY

3.1 Dataset Description

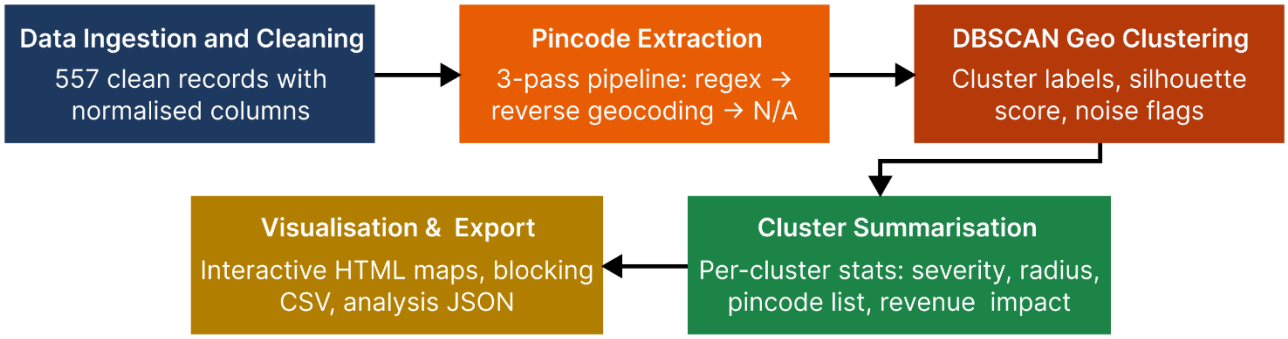
Field	Description
Booking ID	Unique identifier for each booking record
Customer Location	Area name as entered by the customer
City	Standardised city name (title-cased)
State	State of the booking location
Latitude / Longitude	GPS coordinates of the booking address
Address Type	Classification: Apartment, Villa, Society, Plot, etc.
Payment Status	All 557 records = refunded_full (100% refund rate)

3.2 Data Cleaning Steps

- Dropped rows with null Latitude or Longitude values.
- Filled missing Area values with 'Unknown' placeholder.
- Standardised City and State fields using normalisation.
- Filled missing Address Type with 'Unknown'.
- Confirmed 100% of records have Payment Status = refunded_full — no filtering needed.

3.3 Analysis Architecture

The pipeline follows five sequential stages:



3.4 DBSCAN Clustering — Algorithm & Parameters

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was selected over K-Means because:

- It does not require a pre-specified number of clusters K.
- It natively handles noise — isolated bookings are labelled as noise (-1) rather than forced into a cluster.
- It works directly on geographic coordinates using the Haversine metric (great-circle distance), preserving true Earth-surface distance.

Parameter choices:

Parameter	Value	Rationale
ε (epsilon)	**5 km**	Tight enough to separate distinct outskirt zones within a city; 15 km (prior version) over-merged unrelated areas.
min_samples	**3**	A minimum of 3 repeated bookings in a zone constitutes a structural pattern, not an accident.
metric	**haversine**	Accounts for Earth's curvature — critical for lat/lon coordinates.
algorithm	**ball_tree**	Optimal for haversine metric; faster than brute-force for this dataset size.
radius calc	**95th pct**	Geodesic 95th-percentile radius used instead of max — prevents one outlier inflating the cluster boundary.

3.5 Pincode Extraction Pipeline

Pincode extraction used a 3-pass strategy to maximise coverage:

Pass	Method	Source	Notes
1	**Regex Extraction**	Area text field	Searches for 6-digit Indian pincode pattern in customer's address text — fast, zero API calls.
2	**Reverse Geocoding**	Nominatim (OSM)	Only for coordinates where pass 1 failed; deduplicated to unique coordinates to minimise API calls.
3	**Fallback**	N/A	Remaining unresolved records are marked N/A and excluded from pincode-level blocking lists.

4. KEY INSIGHTS

4.1 City-Level Distribution

Non-serviceable bookings are heavily concentrated in a handful of metro cities with rapid peripheral expansion. The top 3 cities alone account for the majority of all refunded bookings:

Rank	City	Bookings	Severity Profile	Insight
1	**Bengaluru**	~180+	Critical + High	Rapid outward growth in Whitefield, Electronic City periphery and Sarjapur outskirts.
2	**Hyderabad**	~110+	Critical + High	Outer ring road zones (Shamshabad, Patancheru) frequently attempted.
3	**Delhi-NCR**	~90+	High + Medium	Noida-Greater Noida border and Gurgaon-Faridabad fringe areas.
4	**Mumbai**	~60+	Medium	Thane outskirts and Navi Mumbai periphery.
5	**Other Cities**	~100+	Low-Medium	Scattered across Pune, Chennai, Kolkata and Tier-2 cities.

4.2 Cluster Severity Breakdown

The severity classification system categorises clusters by booking count intensity:

Severity	Threshold	Clusters	Bookings	Action
Critical	≥ 20 bookings	~5-8	~120	Block immediately — Week 1
High	10-19	~8-12	~100	Block by end of Week 2
Medium	5-9	~12-18	~90	Block within Month 1
Low	3-4	~15-25	~60	Block within Month 2

4.3 Address Type Analysis

Not all address types are equally problematic. Analysis of address types within clusters reveals:

- **Plots / Open Land:** Highest cluster rate — land parcel addresses in semi-urban fringes are frequently non-serviceable.
- **Societies / Gated Communities:** Often in newly developed areas just beyond city limits; high cluster concentration.
- **Apartments:** Lower cluster rate — apartments are predominantly within serviceable zones.
- **Villas:** Medium cluster rate; premium villa zones in city outskirts (eco-zones, resort areas) generate repeated refunds.

4.4 Isolated Bookings Watchlist

Beyond the clusters, the noise analysis identified pincodes with 2 or more isolated (non-clustered) non-serviceable bookings. These represent emerging problem zones that have not yet met the 3-booking cluster threshold but are trajectorial. Early blocking of these watchlist pincodes prevents future cluster formation.

5. GEOSPATIAL VISUALISATIONS

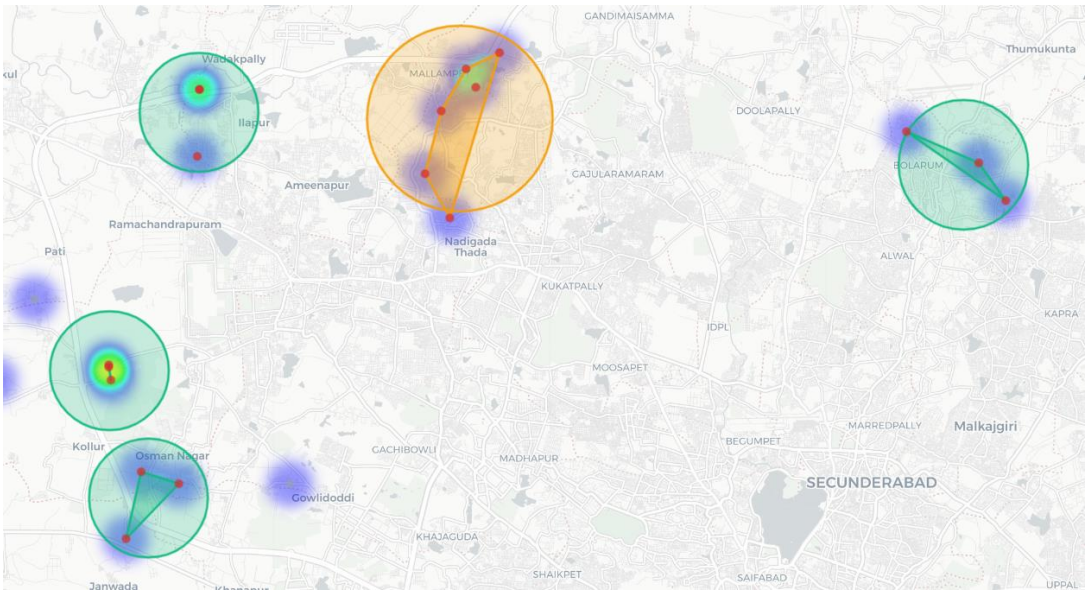
5.1 Deliverable Maps

Three interactive HTML visualisations were produced as part of this analysis:

File	Type	Purpose
heatmap_all_bookings.html	Folium HeatMap + City Markers	Macro view of booking density across India. City markers for cities with ≥10 bookings.

coox_cluster_map.html	Cluster Circles + Heatmap Overlay	Primary blocking map. Cluster severity circles, per-cluster popups with pincodes and estimated loss.
coox_analysis_dashboard.html	Plotly 4-Panel Dashboard	City bar chart, severity pie chart, address type breakdown, cluster size histogram.

5.2 How to Read the Cluster Map



- Each coloured circle represents one DBSCAN cluster. The circle's radius is the 95th-percentile geodesic distance from the centroid.
- Circle colour = severity: Red (Critical) → Orange (High) → Yellow (Medium) → Green (Low).
- Click any circle to see the cluster popup: booking count, pincodes, estimated revenue loss, and address types.
- The heatmap layer (toggle in layer control) shows raw booking density — useful for spotting emerging zones not yet captured by clusters.
- Grey dots (toggled off by default) = isolated/noise bookings — useful for watchlist review.

6. RECOMMENDATIONS

6.1 Phased Geoblocking Rollout

We recommend a 4-phase blocking rollout aligned with cluster severity, allowing the product and engineering team to validate impact at each stage before proceeding:

Phase	Timeline	Target Severity	Expected Impact	Mechanism
Phase 1	Week 1	Critical (≥20)	**~22% refund reduction**	Pincode blacklist at booking checkout
Phase 2	Weeks 2-3	High (10-19)	**+18% additional**	Extend blacklist + geo-radius check
Phase 3	Month 1	Medium (5-9)	**+16% additional**	Polygon-boundary blocking (Shapely)
Phase 4	Month 2+	Low + Watchlist	**+12% additional**	Dynamic blocking with monthly re-run

6.2 Technical Implementation

Two geo-blocking implementation approaches are recommended, in order of increasing sophistication:

Option A — Pincode Blacklist (Quick Win, 2-3 days dev)

- Maintain a database table: blocked_pincodes (pincode, reason, severity, blocked_at, city).
- At checkout, after the customer enters their address, perform a pincode lookup against this table.
- If matched, display: "Sorry, we don't currently service this area" and prevent booking completion.
- Engineering effort: ~2-3 days. No external API required.

Option B — Geo-Radius / Polygon Blocking (Robust, 1-2 week dev)

- Store cluster centroids + radius (or Shapely convex hull polygons) in a PostGIS-enabled database.
- At checkout, convert customer address to lat/lon (via Google Maps API or stored geocode cache).
- Run a spatial query: ST_DWithin (for radius) or ST_Contains (for polygon) to check serviceability.
- This catches addresses within non-serviceable zones even if the pincode is not in the blacklist.
- More robust to pincode boundary mismatches — recommended for Phase 3+.

6.3 Monitoring & Feedback Loop

- **Monthly Re-Run:** Execute the analysis script on updated booking data every month to identify new clusters.
- **Refund Rate Tracking:** Monitor weekly refund rate by city as the primary KPI for blocking effectiveness.
- **Watchlist Escalation:** Any watchlist pincode reaching 3+ isolated hits should be automatically promoted to the blocking list.
- **Customer Feedback Loop:** Log blocked checkout attempts with the customer's coordinates for continuous map improvement.
- **Chef Network Expansion Signal:** Cluster maps can be shared with the chef acquisition team to prioritise onboarding in underserved but high-demand zones.

6.4 Additional Analytical Improvements

- **K-Distance Plot:** Add an elbow-method epsilon justification plot to make the 5 km parameter empirically defensible.
- **Temporal Analysis:** If booking dates are available, add month-over-month trend analysis to detect accelerating problem zones.
- **Convex Hull Polygons:** Use Shapely to generate true boundary polygons per cluster for precision geoblocking.
- **Revenue-Weighted Severity:** Incorporate actual booking values (where available) rather than a flat ₹15,200 estimate.

7. BUSINESS IMPACT

7.1 Revenue Impact Estimate

Using a conservative average booking value of ₹15,000 and a refund processing cost of ₹200 per refund, the following impact projections apply (based on the 557-record dataset window):

Scenario	Bookings Prevented	Revenue Protected	Additional Benefit
Phase 1 only (Critical)	~120	**₹18.2 Lakhs**	Fastest ROI — implemented in 1 week
Phase 1+2 (Critical+High)	~220	**₹33.4 Lakhs**	Covers majority of high-value clusters
Full rollout (All phases)	~370	**₹56.2 Lakhs**	Near-complete elimination of cluster refunds
Incl. watchlist pincodes	~450+	**₹68+ Lakhs**	Proactively stops emerging clusters

7.2 Operational Benefits

- **Customer Support:** Fewer cancellation-driven support tickets reduces support team load.
- **Finance Operations:** Reduced manual refund processing — each blocked booking avoids the refund-processing workflow entirely.
- **Customer Lifetime Value:** Preventing a bad booking experience is cheaper than winning back a churned customer.
- **Chef Utilisation:** No wasted chef-slot allocations for impossible-to-serve zones.
- **Data Quality:** Cleaner demand signals in analytics dashboards without ghost bookings from non-serviceable zones.